

NC words and example part programmes

Numerical Control and Computer Numerical Control

NC words

Following is a list of the different types of words in the formation of a block. Not every NC machine uses all the words. Also, the manner in which the words are expressed will differ between machines. By convention, the words in a block are given in the following order:

SEQUENCE NUMBER (n-words): This is used to identify the block.

PREPARATORY WORD (g-words): This word is used to prepare the controller for instructions that are to follow. For example, the word g02 is used to prepare the NC controller unit for circular interpolation along an arc in the clockwise direction. The preparatory word is needed so that the controller can correctly interpret the data that follow it in the block.

COORDINATES (x-, y-, and z-words): These give the coordinate positions of the tool. In a two-axis system, only two of the words would be used. In a four- or five-axis machine, additional a-words and/or b-words would specify the angular positions.

Although different NC systems use different formats for expressing a coordinate, we will adopt the convention of expressing it in the familiar decimal form: For example, x+7.235 or y-0.500. Some formats do not use the decimal point in writing the coordinate. The + sign to define a positive coordinate location is optional. The negative sign is, of course, mandatory.

FEED RATE (f-word): This specifies the feed in a machining operation. Units are inches per minute (ipm) by convention.

CUTTING SPEED (s-word): This specifies the cutting speed of the process, the rate at which the spindle rotates.

TOOL SELECTION (t-word): This word would be needed only for machines with a tool turret or automatic tool changer. The t-word specifies which tool is to be used in the operation. For example, t05 might be the designation of a 1/2-in. drill bit in turret position 5 on an NC turret drill.

MISCELLANEOUS FUNCTION (m-word): The m-word is used to specify certain miscellaneous or auxiliary functions which may be available on the machine tool. Of course, the machine must possess the function that is being called. An example would be m03 to start the spindle rotation. The miscellaneous function is the last



G & M Codes

Mill G Codes

G00	Rapid Positioning Motion (X,Y,Z,A,B)
G01	Linear Interpolation Motion (X,Y,Z,A,B,F)
G02	Circular Interpolation Motion CW (X,Y,Z,A,I,J,K,R,F)
G03	Circular Interpolation Motion CCW (X,Y,Z,A,I,J,K,R,F)
G04	Dwell (P) (P=Seconds". Milliseconds)
G09	Exact Stop, Non-Modal
G17*	Circular Motion XY Plane Selection (G02 or G03)
G18	Circular Motion ZX Plane Selection (G02 or G03)
G19	Circular Motion YZ Plane Selection (G02 or G03)
G20	Inch Coordinate Positioning
G21	Metric Coordinate Positioning
G28	Machine Zero Return Thru Ref. Point (X,Y,Z,A,B)
G29	Move to Location Through G28 Ref. Point (X,Y,Z,A,B)
G40*	Cutter Comp Cancel G41/G42/G141 (X,Y)
G41	2D Cutter Compensation, Left (X,Y,D)
G42	2D Cutter Compensation, Right (X,Y,D)
G43	Tool Length Compensation + (H,Z)
G49*	Tool Length Compensation Cancel G43/G44/G43
G52	Work Offset Positioning Coordinate
G53	Machine Positioning Coordinate, Non-Modal (X,Y,Z,A,B)
G54*	Work Offset Positioning Coordinate #1 (Setting 56)
G55	Work Offset Positioning Coordinate #2
G56	Work Offset Positioning Coordinate #3
G57	Work Offset Positioning Coordinate #4
G58	Work Offset Positioning Coordinate #5
G59	Work Offset Positioning Coordinate #6

G73	HS Peck Drilling Canned Cycle (X,Y,A,B,Z,I,J,K,Q,P,R,L,F)
G74	Reverse Tapping Canned Cycle (X,Y,A,B,Z,J,R,L,F)
G76	Fine Boring Canned Cycle (X,Y,A,B,Z,I,J,P,Q,R,L,F)
G77	Black Bore Canned Cycle (X,Y,A,B,Z,I,J,Q,R,L,F)
G80*	Cancel Canned Cycle (Setting 56)
G81	Drill Canned Cycle (X,Y,A,B,Z,R,L,F)
G82	Spot Drill / Counterbore Canned Cycle (X,Y,A,B,Z,P,R,L,F)
G83	Peck Drill Deep Hole Canned Cycle (X,Y,A,B,Z,I,J,K,Q,P,R,L,F)
G84	Tapping Canned Cycle (X,Y,A,B,Z,J,R,L,F)
G85	Bore In ~ Bore Out Canned Cycle (X,Y,A,B,Z,R,L,F)
G86	Bore In ~ Stop ~ Rapid Out Canned Cycle (X,Y,A,B,Z,R,L,F)
G87	Bore In ~ Manual Retract Canned Cycle (X,Y,A,B,Z,R,L,F)
G88	Bore In ~ Dwell ~ Manual Retract Canned Cycle (X,Y,A,B,Z,P,R,L,F)
G89	Bore In ~ Dwell ~ Bore Out Canned Cycle (X,Y,A,B,Z,P,R,L,F)
G90*	Absolute Positioning Command
G91	Incremental Positioning Command
G92	Global Work Coordinate System
G93	Inverse Time Feed Mode ON
G94	Inverse Time Feed OFF / Feed Per Minute ON
G98	Canned Cycle Initial Point Return
G99	Canned Cycle R Plane Return

Mill M Codes

M00	Program Stop
M01	Optional Program Stop
M02	Program End
M03	Spindle ON Clockwise (S)
M04	Spindle ON Counterclockwise (S))
M05	Spindle Stop
M06	Tool Change (T)
M08	Coolant ON
M09	Coolant OFF
M30	Program End and Reset
M31	Chip Auger Forward
M33	Chip Auger Stop
M34	Coolant Spigot Position Down, Increment
M35	Coolant Spigot Position Up, Decrement
M36	Pallet Part Ready
M41	Spindle Low Gear Override
M42	Spindle High Gear Override
M50	Execute Pallet Change
M83	Auto Air Jet ON
M84	Auto Air Jet OFF
M88	Coolant Through Spindle ON
M97	Local Sub-Program Call (P , L)
M98	Sub-Program Call (P , L)
M99	Sub-Program / Routine Return of Loop (P)

Notes:

[Cutting tool diameter = 20]

N01 G71 G90 G94

N02 G00 X-20 Y0

N03 G00 Z10

N04 G01 G42 D20 Z-10 S1000 F200 M03

N05 G01 X0 Y0

N06 G01 X90

N07 G01 Y90

N08 G01 X80 Y100

N09 G01 X10 Y

N10 G03 X0 Y90 R10

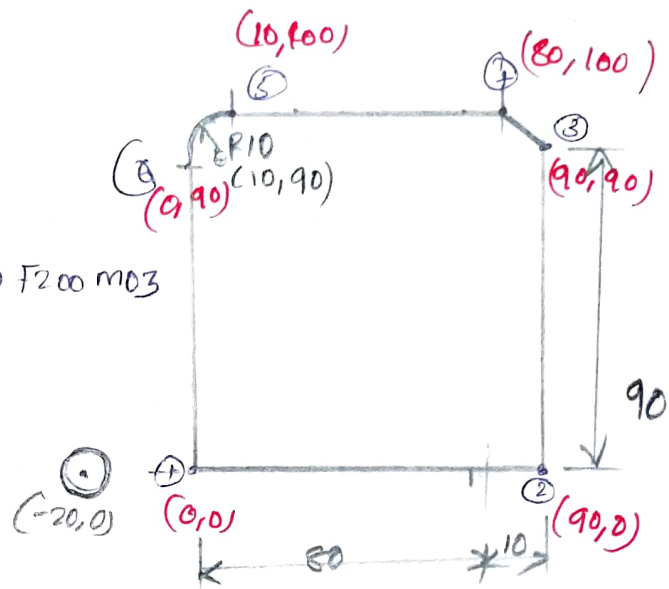
N11 G01 Y0

N12 G40

N13 G00 X-20 Y0

N14 G00 Z0

N15 M30



(X0 Y90 I0 J-10)

Q28: machine zero (Home Point)

481 - Drilling cycle

480 - Drilling cycle comm.)

499 - Repeat Point return

581 X-Y-Z-R-F-

NOI 600 ^{Repeat point} 628

NO2 MO6 TO3 (Drill bit dia 10mm)

3 holes dia. 10mm
through

N03 G90 G80 X10 Y10 Z10 S1100 M03 M08

N04 699 681 X10 Y10 Z-17 R2 F100

N05 $\times 30^{40} 40$

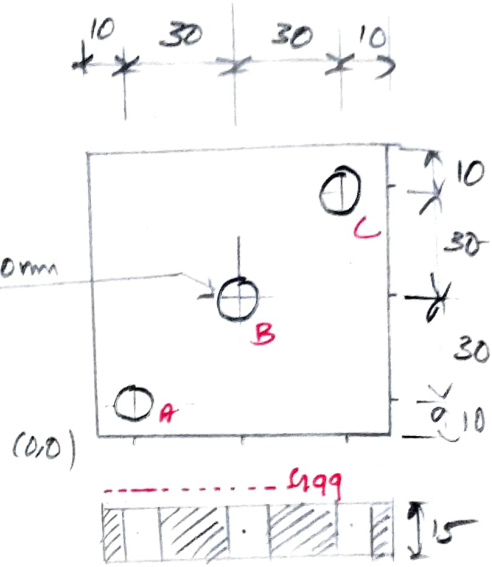
206 ⁷⁰~~40~~ ⁷⁰Y40

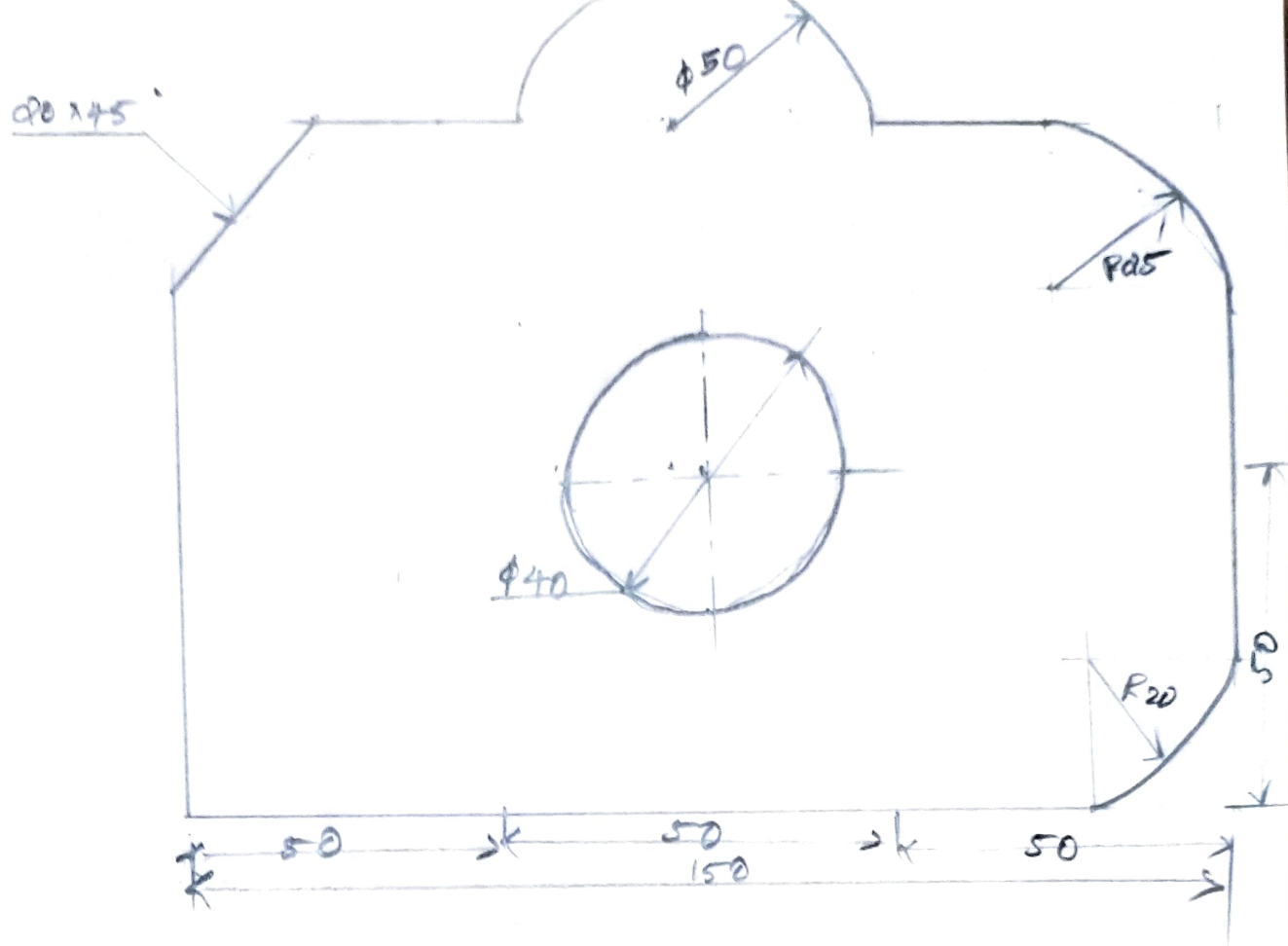
NO7 900 980 210

NO8 400 mos moq

noq 6100 628

N10 M30





N01 G00 G28

N02 M06 T03 (end mill dia 10mm)

N03 G90 G00 X0 Y0 Z5 S900 M03 M08

N04 G01 G42 D10 X0 Y0 Z-30 F100

N05 X130

N06 G03 X150 Y20 R20

N07 G01 ~~X150~~ Y75

N08 G03 X125 Y100 R25

N09 G01 X100

N10 G03 X50 Y100 R25

N11 G01 X20 Y100

N12 G01 X0 Y80

N13 G01 X0 Y0

N14 G00 G40 Z5 M05 M09

N15 G00 G28

N16 M01 optional stop

N17 M06 T04 Drill bit dia 40mm

N18 G90 G00 X75 Y50 Z5 S900 M03 M08

N19 G83 X75 Y50 Z-32 Q4 R2 F80

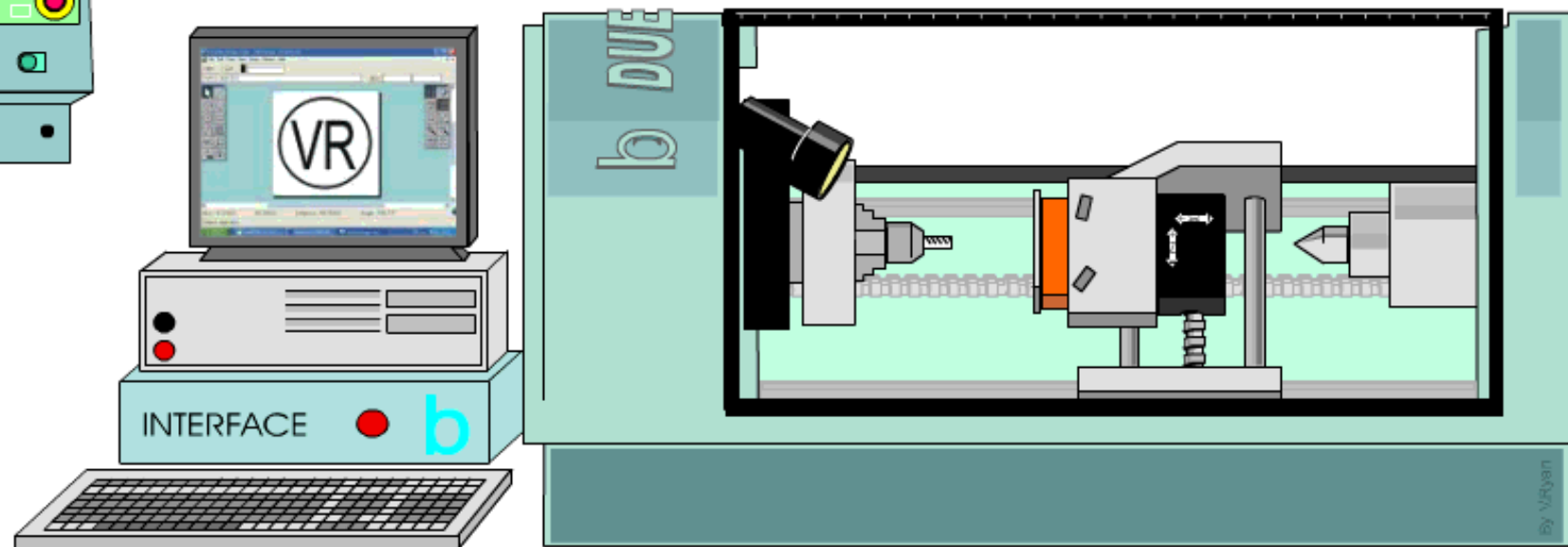
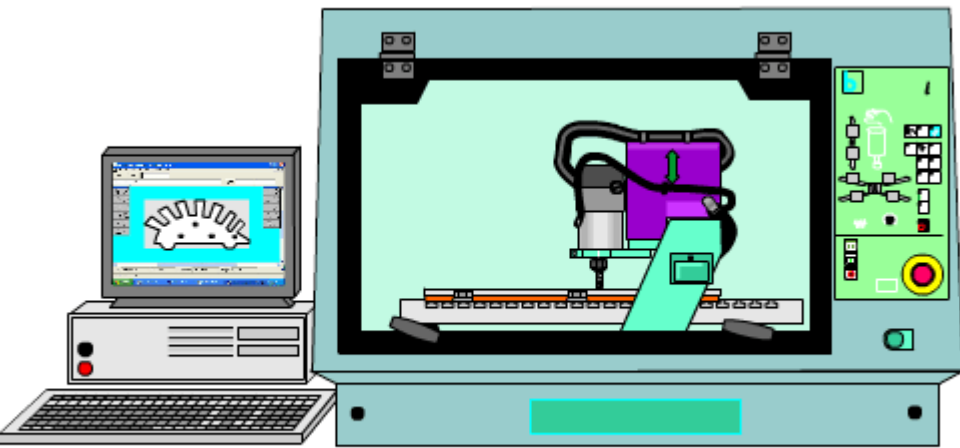
N20 G00 G80 Z5 M05 M09

N21 G00 G28

N22 M30

Numerical Control

COMPUTER NUMERICAL CONTROL
C N C



Numerical Control

- a form of programmable automation in which the process is controlled by numbers, letters, and symbols
- a program of instructions designed for a particular work-part or job
- When the job changes, the program of instructions is changed.
- This capability to change the program for each new job is what gives NC its flexibility

BASIC COMPONENTS OF AN NC SYSTEM

1. Program of instructions
2. Controller unit, also called a machine control unit (MCU)
3. Machine tool or other controlled process

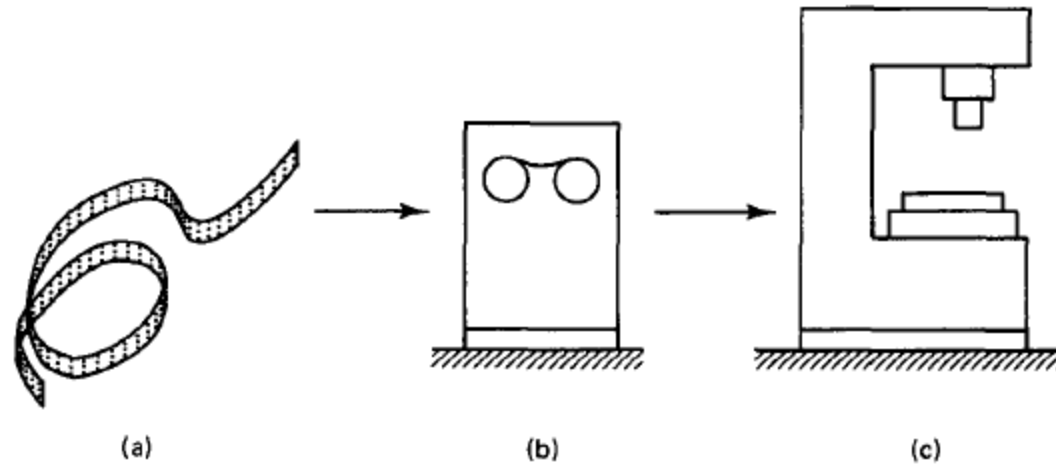


FIGURE 7.1 Three basic components of a numerical control system: (a) program of instruction; (b) controller unit; (c) machine tool.

1. Program of instructions

- The program of instructions is the detailed step-by-step set of directions which tell the machine tool what to do.
- It is coded in numerical or symbolic form on some type of input medium that can be interpreted by the controller unit.
- *The part program is converted into a sequence of machine tool actions by means of the input medium, which contains the program the controller unit, which interprets the input medium.*
- The controller unit and the input medium must be compatible - the input medium uses coded symbols which represent the part program, and the controller unit must be capable of reading those symbols

PUNCHED TAPE IN NC

- There are eight regular columns of holes running in the lengthwise direction of the tape.
- There is also a ninth column of holes between the third and fourth regular columns. However, these are smaller and are used as sprocket holes for feeding the tape.
- The coding of the tape is provided by either the presence or absence of a hole in the various positions.

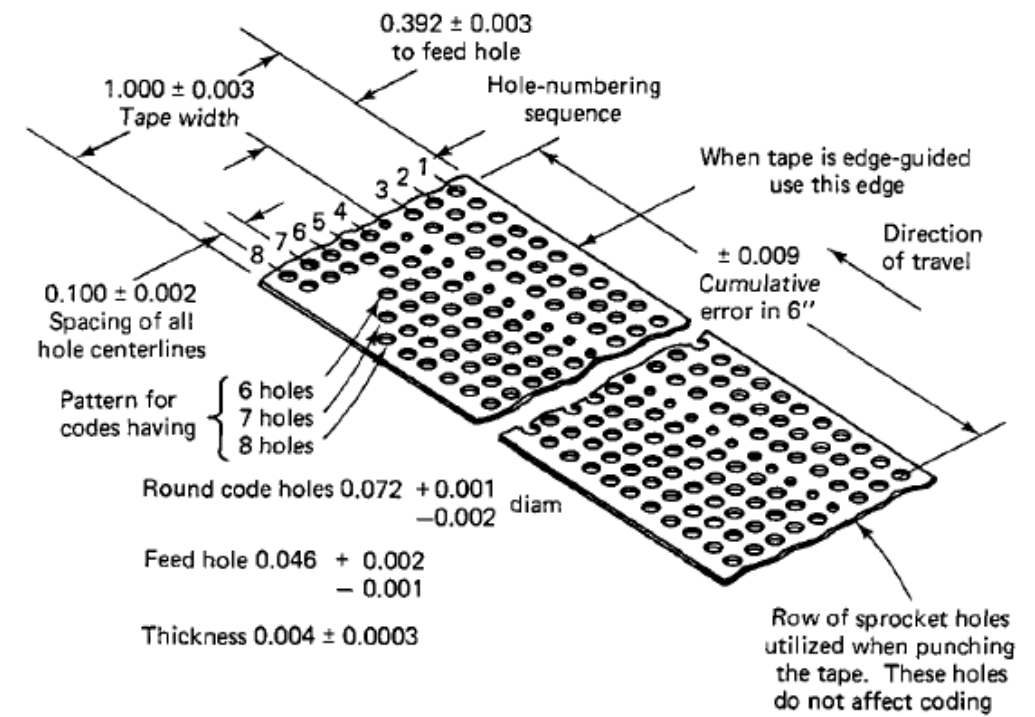
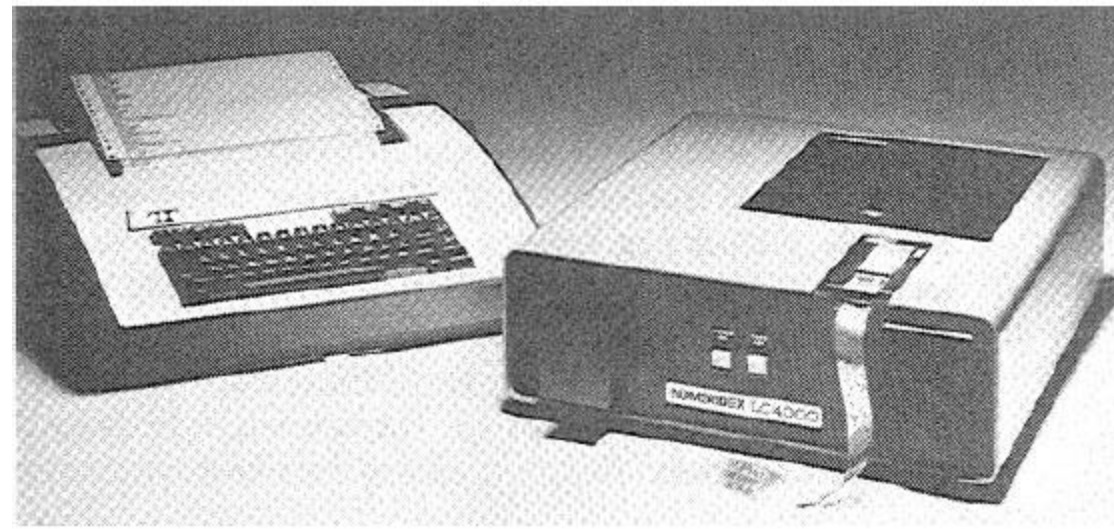


FIGURE 8.1 Numerical control punched tape format as standardized by Electronic Industries Association. (Reprinted from Childs [2].)

- Because there are two possible conditions for each position—either the presence or absence of a hole—this coding system is called the binary code.
- The NC tape coding system is used to code not only numbers, but also alphabetical letters and other symbols.
- Eight columns provide more than enough binary digits to define any of the required symbols.
- A character is a combination of bits, which represents a letter, number, or other symbol.
- A word is a collection of characters used to form part of an instruction.

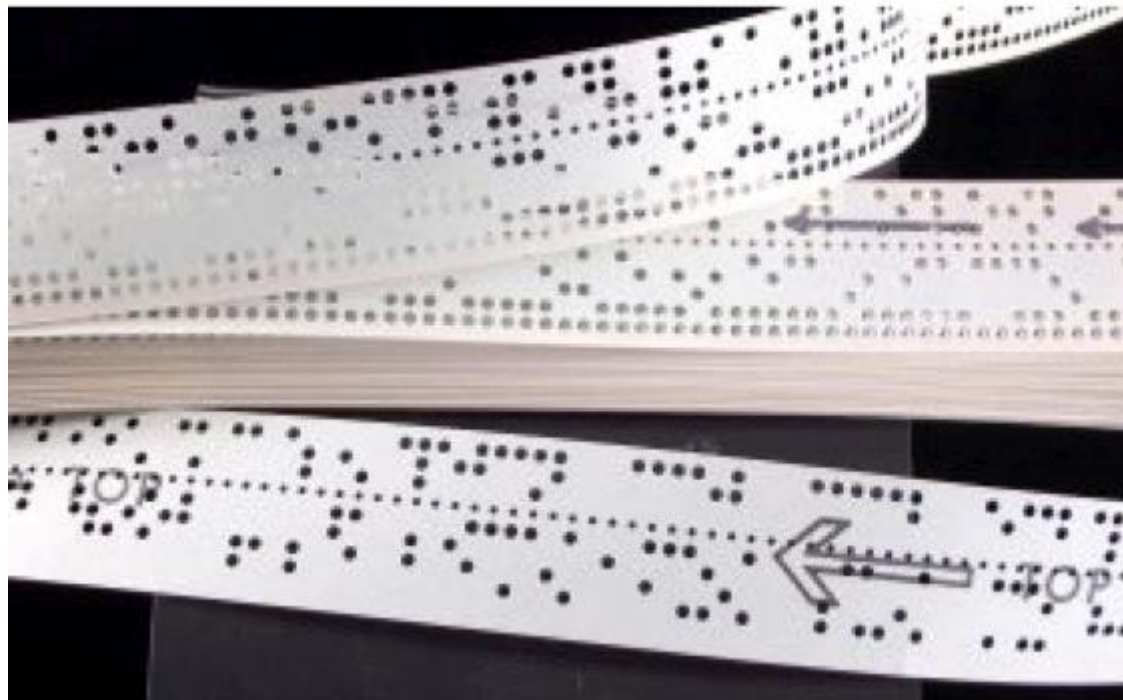
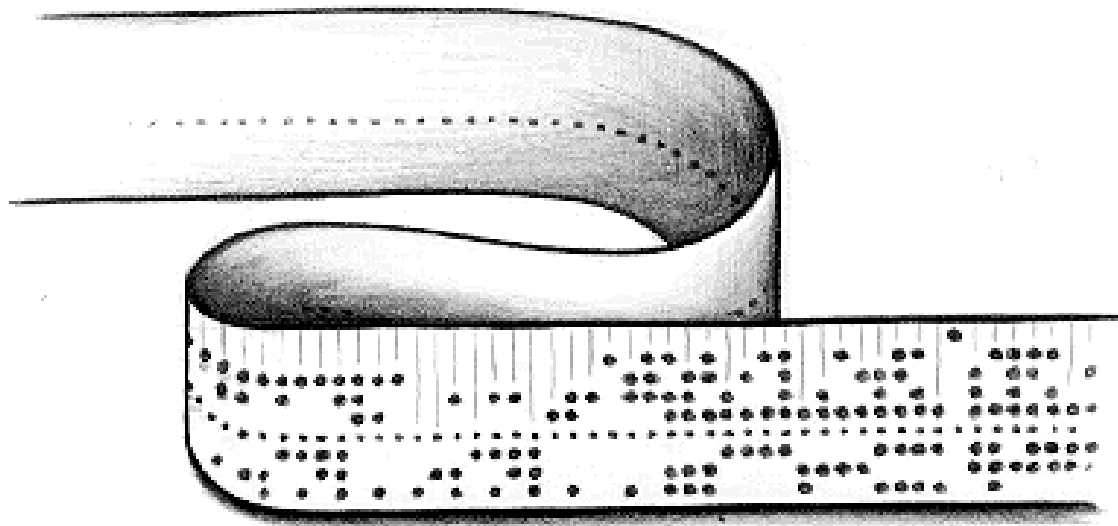
Preparing the punched tape

- There are two basic methods of preparing the punched tape.
- The first method is associated with manual part programming and involves the use of a typewriterlike device.
- After the last instruction has been read into the controller, the tape is rewound back to the start of the program to be ready for the next work-part.



8.2 Equipment used to prepare punched tape for NC. (Courtesy of Numeridex Corp.)

- The second method is used with computer-assisted part programming - by this approach, the tape is prepared directly by the computer using a device called a tape punch.
- During production on a conventional NC machine, the tape is fed through the tape reader once for each workpiece.
- It is advanced through the tape reader one instruction at a time.
- While the machine tool is performing one instruction, the next instruction is being read into the controller unit's data buffer.
- This makes the operation of the NC system more efficient.



Binary Symbol Table

A	□■□□ □□■	a	□■□□ □□■	0	□□■□ □□□□
B	□■□□ □□□	b	□■□□ □□■	1	□□■□ □□□■
C	□■□□ □□■	c	□■□□ □□■	2	□□■□ □□□□
D	□■□□ □■□□	d	□■□□ □□□□	3	□□■□ □□□■
E	□■□□ □□■	e	□■□□ □□■	4	□□■□ □■□□
F	□■□□ □■□□	f	□■□□ □■□□	5	□□■□ □■□■
G	□■□□ □■□■	g	□■□□ □■□■	6	□□■□ □■□□
H	□■□□ ■□□□	h	□■□□ ■□□□	7	□□■□ □■□■
I	□■□□ ■□□■	i	□■□□ ■□□■	8	□□■□ ■□□□
J	□■□□ ■□□□	j	□■□□ ■□□□	9	□□■□ ■□□■
K	□■□□ □□■	k	□■□□ ■□□■	.	□□□□ ■□□□
L	□■□□ ■□□□	l	□■□□ ■□□□	,	□□□□ ■□□□
M	□■□□ ■□□■	m	□■□□ ■□□■	:	□□■□ ■□□□
N	□■□□ ■□□□	n	□■□□ ■□□□	;	□□■□ □□■□
O	□■□□ ■□□■	o	□■□□ ■□□■	?	□□■□ ■□□■
V	□■□■ □□□□	p	□■□■ □□□□	!	□□■□ □□□■
Q	□■□■ □□□■	q	□■□■ □□□■	#	□□■□ □□■□
R	□■□■ □□□□	r	□■□■ □□□□	&	□□■□ □□■□
S	□■□■ □□□■	s	□■□■ □□□■	'	□□■□ □□■□
T	□■□■ □■□□	t	□■□■ □■□□	"	□□■□ □□■□
U	□■□■ □□□■	u	□■□■ □□□■	-	□□■□ ■□□■
V	□■□■ □■□□	v	□■□■ □■□□		
W	□■□■ □■□■	w	□■□■ □■□■		
X	□■□■ ■□□□	x	□■□■ ■□□□		
Y	□■□■ ■□□■	y	□■□■ ■□□■		
Z	□■□■ ■□□■	z	□■□■ ■□□■		

2. Controller unit

- The second basic component of the NC system is the controller unit.
- This consists of the electronics and hardware that read and interpret the program of instructions and convert it into mechanical actions of the machine tool.
- The typical elements of a conventional NC controller unit include
 - The tape reader
 - a data buffer
 - Signal output channels to the machine tool
 - Feedback channels from the machine tool
 - The sequence controls to coordinate the overall operation of the foregoing elements.

- The tape reader is an electromechanical device for winding and reading the punched tape containing the program of instructions.
- The data contained on the tape are read into the data buffer
- The purpose of this device is to store the input instructions in logical blocks of information.
- A block of information usually represents one complete step in the sequence of processing elements.
 - For example, one block may be the data required to move the machine table to a certain position and drill a hole at that location.
- The signal output channels are connected to the servomotors and other controls in the machine tool.
- Through these channels, the instructions are *sent to the machine tool from the controller unit*
- To make certain that the instructions have been properly executed by the machine, feedback data are sent back to the controller via the feedback channels.
- The most important function of this return loop is to assure that the table and work-part have been properly located with respect to the tool.
- Sequence controls coordinate the activities of the other elements of the controller unit.
- The tape reader is actuated to read data into the buffer from the tape, signals are sent to and from the machine tool, and so on. - These types of operations must be synchronized and this is the function of the sequence controls.

3. Machine tool

- The third basic component of an NC system is the machine tool or other controlled process.
- It is capable of performing a variety of different operations: drilling, tapping, reaming, milling, and boring
- A variety of machining operations means that a variety of cutting tools are required.
- The tools are kept in a tool drum or other holding device. When the tape calls a particular tool, the drum rotates to position the tool for insertion into the spindle.
- The automatic tool changer then grasps the tool and places it into the spindle chuck.

NC Procedure

To utilize numerical control in manufacturing, the following steps must be accomplished.

1. *Process planning.*

- The engineering drawing of the work-part must be interpreted in terms of the manufacturing processes to be used.
- This step is referred to as process planning and it is concerned with the preparation of a **route sheet**.
- The route sheet is a listing of the sequence of operations which must be performed on the work-part.
- It is called a route sheet because it also *lists the machines through which the part must be routed in order to accomplish the sequence of operations*

2. *Part programming*

- Manual part programming
- Computer-assisted part programming

3. *Tape preparation.*

- A punched tape is prepared from the part programmer's NC process plan

4. *Tape verification*

5. *Production*

Absolute positioning and incremental positioning

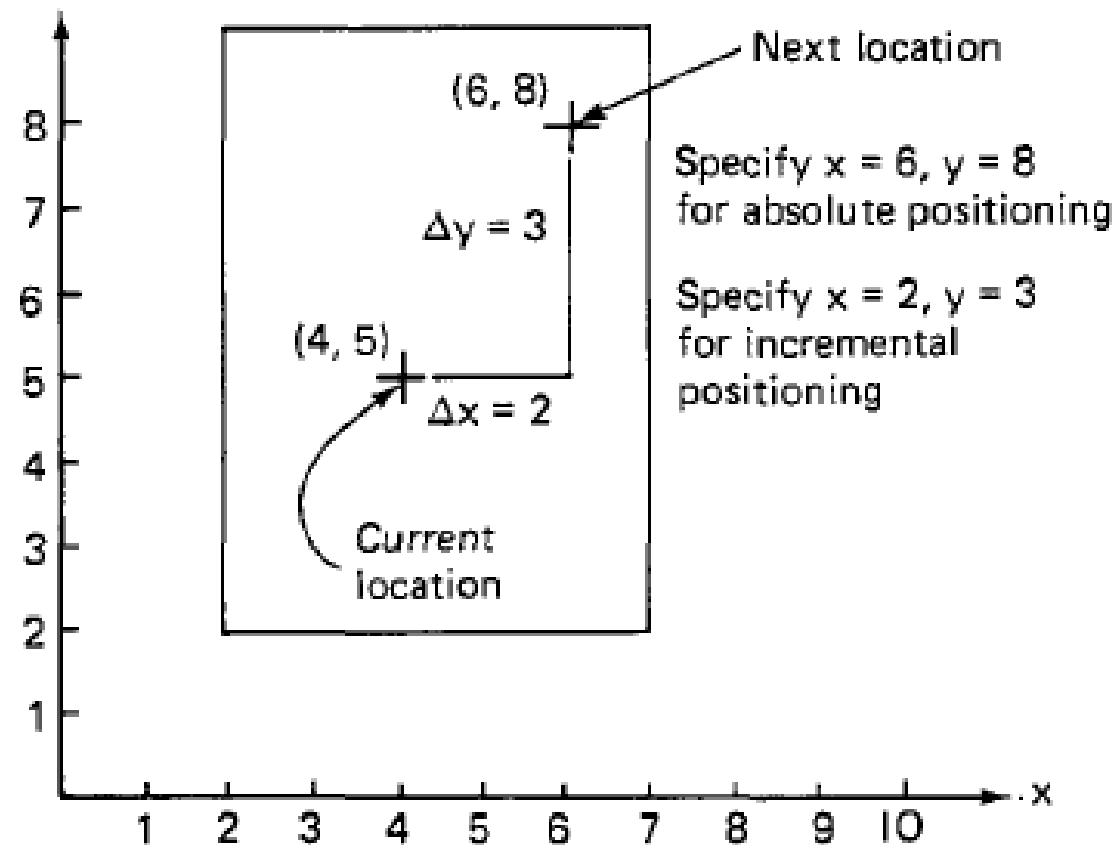


FIGURE 7.6 Absolute versus incremental positioning

NC MOTION CONTROL SYSTEMS

In order to accomplish the machining process, the cutting tool and workpiece must be moved relative to each other

There are three basic types of motion control systems:

Point-to-point NC:

- In PTP, the objective of the machine tool control system is to move the cutting tool to a predefined location
- The speed or path by which this movement is accomplished is not important
- Once the tool reaches the desired location, the machining operation is performed at that position.
- NC drill is a good example of PTP systems

Straight-cut NC

- Straight-cut control systems are capable of moving the cutting tool parallel to one of the major axes at a controlled rate suitable for machining
 - Appropriate for performing milling operations
 - it is not possible to combine movements in more than a single axis direction.
- Therefore, angular cuts on the workpiece would not be possible.
- An example of a-straight-cut operation is shown in Figure

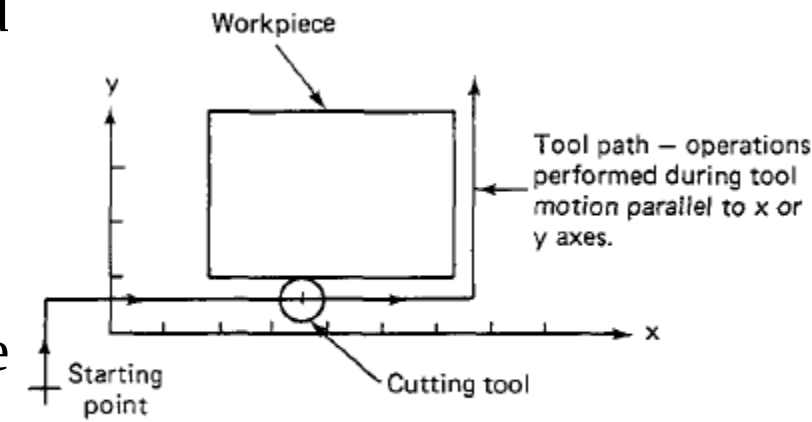


FIGURE 7.8 Straight-cut system.

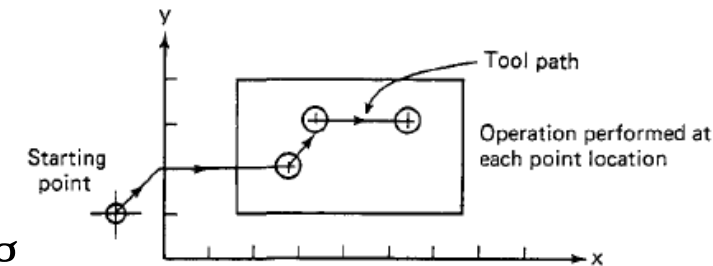


FIGURE 7.7 Point-to-point (positioning) NC system.

NC MOTION CONTROL SYSTEMS

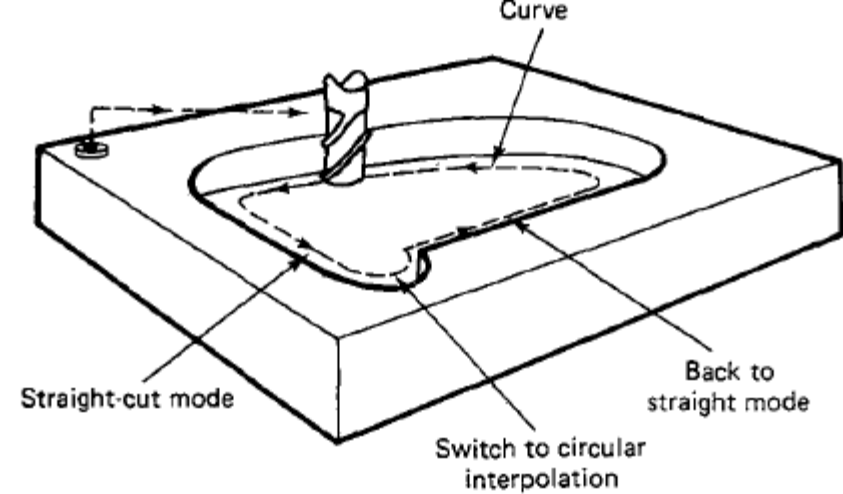


FIGURE 7.9 Contouring (continuous path) NC system for two-dimensional operations. (Reprinted from Olesten [9].)

Contouring NC:

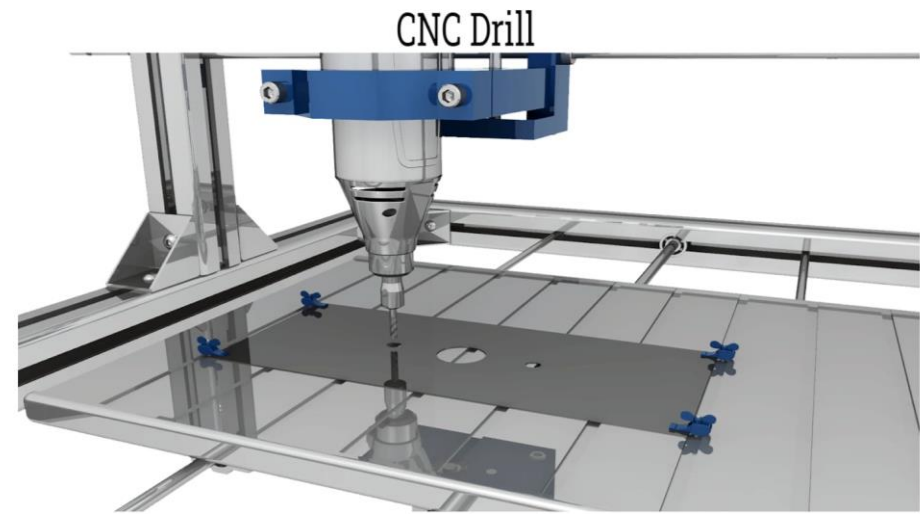
- Contouring is the most complex, the most flexible, and the most expensive type of tool control
- It is capable of performing both PTP and straight-cut operations
- In addition, simultaneous control of more than one axis movement of the machine tool
- The path of the cutter is continuously controlled to generate the desired geometry of the workpiece
- For this reason, contouring systems are also called continuous-path NC systems.
- Straight or plane surfaces at any orientation, circular paths, conical shapes are possible under contouring control

Advantages of NC

- 1. Reduced nonproductive time*
- 2. Reduced fixturing*
- 3. Reduced manufacturing lead time*
- 4. Greater manufacturing flexibility*
- 5. Improved quality control*
- 6. Reduced inventory*
- 7. Reduced floor space requirement*

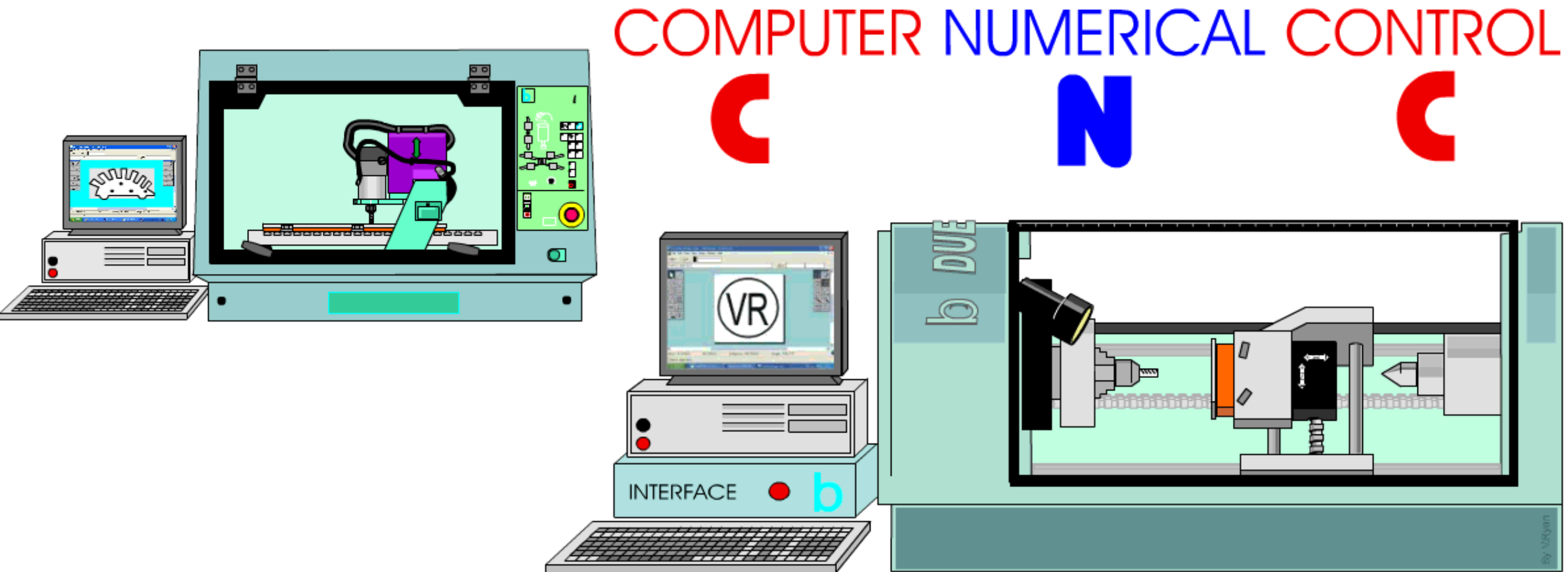
Disadvantages of NC

- 1. Higher investment*
- 2. Higher maintenance cost*
- 3. Finding and/or training NC personnel*



Computer Controls in NC

The computer is being used to refine and improve the NC part programming procedure through interactive graphics and voice programming



PROBLEMS WITH CONVENTIONAL NC

1. *Part programming mistakes.*
2. *Nonoptimal speeds and feeds* - In conventional numerical control, the control system does not provide the opportunity to make changes in speeds and feeds during the cutting process
3. *Punched tape* - Paper tape is especially fragile, and its susceptibility to wear and tear causes unreliable NC component for repeated use in the shop
4. *Tape reader* - least reliable hardware component of the machine
5. *Controller.* The conventional NC controller unit is hard-wired, control features cannot be easily altered to incorporate improvements into the unit - Use of a computer as the control device would provide the flexibility to make improvements (becomes software)
6. *Management information.* The conventional NC system is not equipped to provide timely information on operational performance to management

COMPUTER NUMERICAL CONTROL

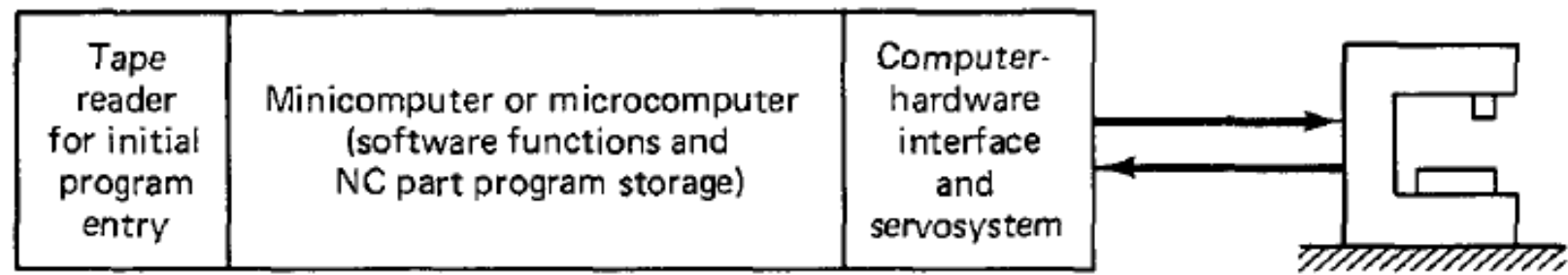


FIGURE 9.1 General configuration of computer numerical control (CNC) system.

- Computer numerical control is an NC system that utilizes a dedicated, stored program computer to perform some or all of the basic numerical control functions.
- Part programs are initially entered in a similar manner.
- Punched tape readers are still the common device to input the part program into the system.
- However, in conventional numerical control, the punched tape is cycled through the reader for every workpiece in the batch.
- With CNC, the program is entered once and then stored in the computer memory.
- Thus the tape reader is used only for the original loading of the part program and data.
- Compared to regular NC, CNC offers additional flexibility and computational capability.
- New system options can be incorporated into the CNC controller simply by reprogramming
- Because of this reprogramming capacity, CNC is often referred to by the term "soft-wired" NC

Functions of CNC

- There are a number of functions which CNC is designed to perform.
- Several of these functions would be either impossible or very difficult to accomplish with conventional NC.

The principal functions of CNC are:

1. Machine tool control
2. In-process compensation
3. Improved programming and operating features
4. Diagnostics

MACHINE TOOL CONTROL

- The primary function of the CNC system is control of the machine tool.
 - This involves conversion of the part program instructions into machine tool motions through the computer interface and servosystem
- The controller consists of the soft-wired controller unit and hard-wired controller unit (logic circuits or data buffer blocks)
 - The capability to conveniently incorporate a variety of control features into the soft-wired controller unit is the main advantage of CNC.
 - Some of the control functions, such as circular interpolation, can be accomplished more efficiently with hard-wired controller unit

Two alternative controller designs in CNC:

1. Hybrid CNC
2. Straight CNC

Hybrid CNC and Straight CNC

Hybrid CNC

- hard-wired controller performs feed rate generation and circular interpolation.
- soft-wired controller performs remaining control functions

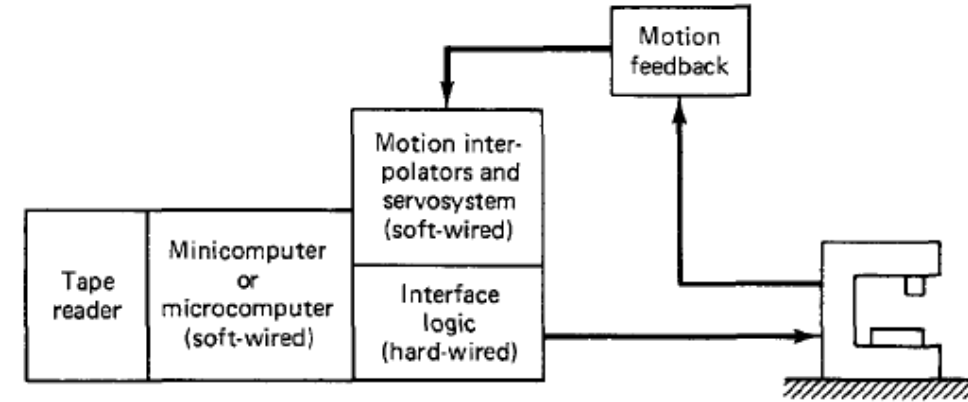


FIGURE 9.2 Hybrid CNC.

Straight CNC

- Computer (soft-wired controller) performs all the NC functions such as Interpolation, tool position feedback
- hard-wired controller required to interface the computer with the machine tool and the operator's console. Interpolation, tool position feedback, and
- the computer required in a straight CNC system must be more powerful than that needed for a hybrid system

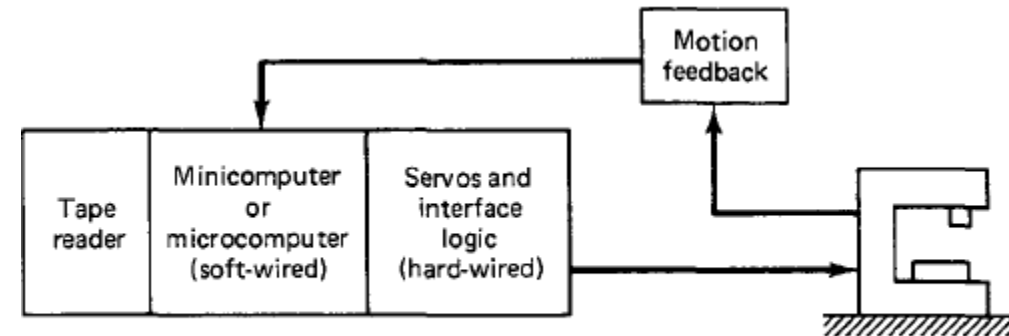


FIGURE 9.3 Straight CNC.

IN-PROCESS COMPENSATION

IN-PROCESS COMPENSATION - dynamic correction of the machine tool motions for changes or errors which occur during processing.

Some of the CNC in-process compensation are:

- Adjustments for errors sensed by in-process inspection
- Re-computation of axis positions when an inspection probe is used to locate a datum reference on a work-part
- Offset adjustments for tool radius and length
- Adaptive control adjustments to speed and/or feed

IMPROVED PROGRAMMING AND OPERATING FEATURES and DIAGNOSTICS

IMPROVED PROGRAMMING AND OPERATING FEATURES - The flexibility of soft-wired control has permitted the introduction of many convenient programming and operating features

- Editing of part programs at the machine.
- Graphic display of the tool path to verify the tape.
- Various types of interpolation: circular, parabolic, and cubic interpolation.
- Support of both U.S. customary units and metric units (International System)

DIAGNOSTICS - NC machine tools are complex and expensive systems

The complexity increases the risk of component failures which lead to system downtime. Therefore avoid downtime as much as possible.

ADVANTAGES OF CNC

- 1. The part program tape and tape reader are used only once*
- 2. Tape editing at the machine site*
- 3. Metric conversion*
- 4. Greater flexibility*
- 5. User-written programs*
- 6. Total manufacturing system*

Computer-assisted Part Programming

- Employ the high-speed digital computer to assist in the part programming process.
- Many part-programming language systems have been developed to perform automatically most of the calculations which the programmer would otherwise be forced to do.
- This saves time and results in a more accurate and more efficient part program.

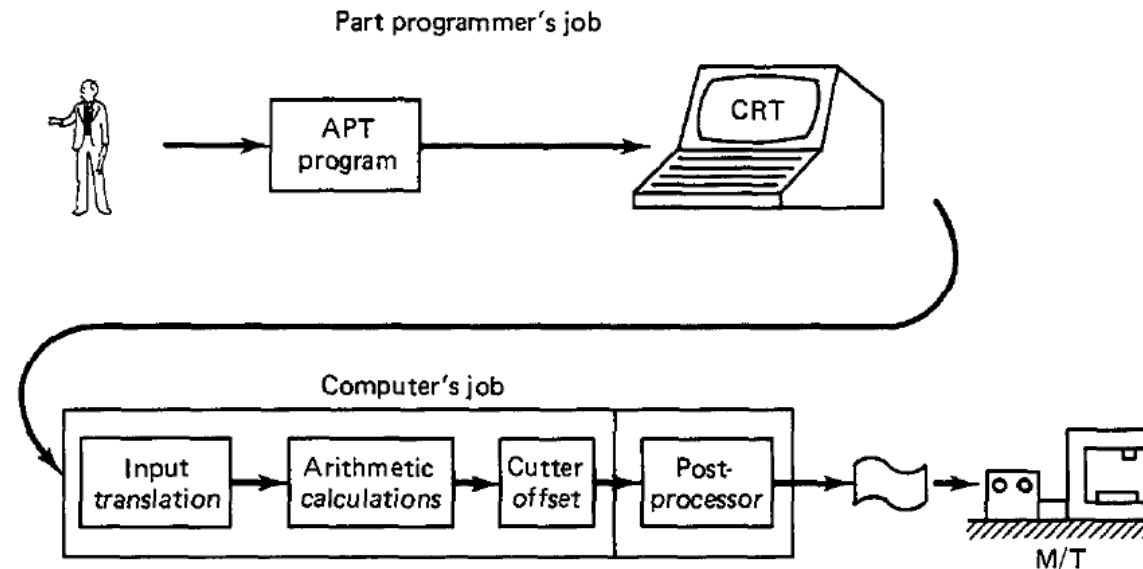


FIGURE 8.7 Steps in computer-assisted part programming.

Computer-assisted Part Programming

The part programmer's job

The part programmer's responsibility in computer-assisted part programming consists of two basic steps:

1. Defining the workpart geometry
 2. Specifying the operation sequence and tool path
- It is the part programmer's task to enumerate the elements out of which the part is composed.
 - Each geometric element must be identified and the dimensions and location of the element explicitly defined.
 - After defining the work-part geometry, the programmer must next construct the path that the cutter will follow to machine the part.
 - This tool path specification involves a detailed step-by-step sequence of cutter moves

Computer-assisted Part Programming

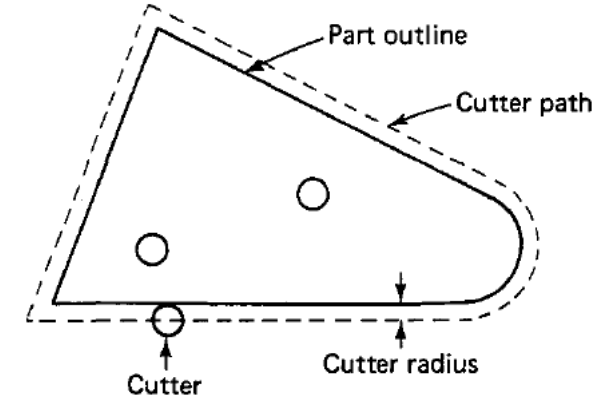
The computer's job

The computer's job in computer-assisted part programming consists of the following steps:

1. Input translation
 2. Arithmetic calculations
 3. Cutter offset computation
 4. Postprocessor
- The part programmer enters the program written in the APT or other language. The input translation component *converts the coded instructions contained in the program into computer-usable form*, preparatory to further processing.
 - The arithmetic calculations unit of the system consists of *a comprehensive set of subroutines for solving the mathematics required to generate the part surface*. Includes time-consuming geometry and trigonometry calculations, to concentrate on the work-part processing.

Computer-assisted Part Programming

- The important task of the part programmer is that of constructing the tool path.
- The actual tool path is different from the part outline because the tool path is defined as the path taken by the center of the cutter.
- The purpose of the **cutter offset computations** is to offset the tool path from the desired part surface by the radius of the cutter. This means that the part programmer can define the exact part outline in the geometry statements.
- NC machine tool systems are different. They have different features and capabilities. They use different NC tape formats.
- The final task of the computer in computer-assisted part programming is to take the general instructions and make them specific to a particular machine tool system.
- The unit that performs this task is called a **postprocessor**.



NC part programming languages

- APT (Automatically Programmed Tools) - The product developed by of the MIT developmental work on NC programming systems in June 1956 and it was first used in production around 1959
- *ADAPT*
- *EXAPT (Extended subset of APT)*
- *UNIAPT*
- *SPLIT (Sundstrand Processing Language Internally Translated)*
- *COMPACT II*
- *PROMPT*
- *CINTURNII*

Reference: Numerical Control, The Beginnings of CAM – CAD/CAM by M. P. Groover

The APT Language

- APT language is English-like statements and to show how they are used to command the cutting tool through its sequence of machining operations.
 1. Geometry statements - These define the geometric elements that comprise the work-part
 2. Motion statements - These are used to describe the path taken by the cutting tool.
 3. Postprocessor statements - They are used to specify feeds and speeds and to actuate other features of the machine.
 4. Auxiliary statements - These are miscellaneous statements used to identify the part, tool, tolerances, and so on.

Geometry statements - *The definition of the work-part elements*

- The general form of an APT geometry statement is this:

symbol = geometry type/descriptive data

PI = POINT/5.0, 4.0, 0.0

The statement is made up of three sections.

- The first is the symbol used to identify the geometric element
- The second section of the geometry statement identifies the type of geometry element - POINT, LINE, PLANE, and CIRCLE.
- The third section of the geometry statement - May include quantitative dimensional and positional data
- A slash separates the surface type from the descriptive data
- Commas are used to separate the words and numbers in the descriptive data

Geometry statements

To specify a line, the easiest method is by two points through which the line passes:

$$L3 = \text{LINE}/P3, P4$$

The part programmer may find it convenient to define a new line parallel to another line which has previously been defined. One way of doing this would be

$$L4 = \text{LINE}/P5, \text{PARLEL}, L3$$

This states that the line L4 must pass through point P5 and be parallel (PARLEL) to line L3.

A plane can be defined by specifying three points through which it passes:

$$PL1 = \text{PLANE}/P1, P4, P5$$

Of course, the three points must not lie along a straight line.

Geometry statements

`PL2 = PLANE/P2, PARLEL, PL1`

Plane PL2 is parallel to plane PL1 and passes through point P2.
A circle can be specified by its center and its radius.

`C1 = CIRCLE/CENTER, P1, RADIUS, 5.0`

The two APT descriptive words are used to identify the center and radius. The orientation of the circle perhaps seems undefined. By convention, it is a circle located in the x - y plane.

1. The coordinate data must be specified in the order x, y, z . For example, the statement

$$P1 = \text{POINT}/5.0, 4.0, 0.0$$

is interpreted by the APT program to mean a point $x = 5.0$, $y = 4.0$, and $z = 0.0$.

2. Any symbols used as descriptive data must have been previously defined. For example, in the statement

$$P2 = \text{POINT}/\text{INTOF}, L1, L2$$

the two lines $L1$ and $L2$ must have been previously defined. In setting up the list of geometry statements, the APT programmer must be sure to define symbols before using them in subsequent statements.

3. A symbol can be used to define only one geometry element. The same symbol cannot be used to define two different elements. For example, the following sequence would be incorrect:

$$\begin{aligned} P1 &= \text{POINT}/1.0, 1.0, 1.0 \\ P1 &= \text{POINT}/2.0, 3.0, 4.0 \end{aligned}$$

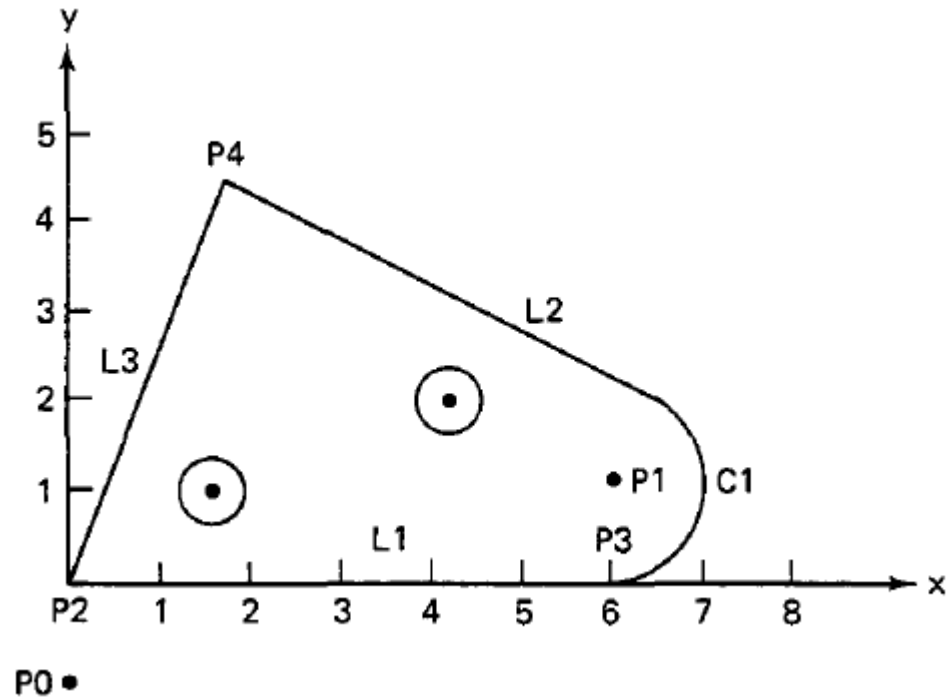
4. Only one symbol can be used to define any given element. For example, the following two statements in the same program would render the program incorrect:

$$\begin{aligned} P1 &= \text{POINT}/1.0, 1.0, 1.0 \\ P2 &= \text{POINT}/1.0, 1.0, 1.0 \end{aligned}$$

5. Lines defined in APT are considered to be of infinite length in both directions. Similarly, planes extend indefinitely and circles defined in APT are complete circles.

Rules that must be followed in formulating an APT geometry statement

Example



P0 = POINT/0, -1.0, 0
P1 = POINT/6.0, 1.125, 0
P2 = POINT/0, 0, 0
P3 = POINT/6.0, 0, 0
P4 = POINT/1.75, 4.5, 0
L1 = LINE/P2, P3
C1 = CIRCLE/CENTER, P1, RADIUS, 1.125
L2 = LINE/P4, LEFT, TANTO, C1
L3 = LINE/P2, P4
PL1 = PLANE/P2, P3, P4

Motion statements

The general form of a motion statement is

motion command/descriptive data

GOTO/PI

The statement consists of two sections separated by a slash.

- The first section is the basic motion command, which tells the tool what to do.
- The second section is comprised of descriptive data, which tell the tool where to go

❖ At the beginning of the motion statements, the tool must be given a starting point - target point is the location where the operator has positioned the tool at the start of the job.

FROM/TARG

or

FROM/-2.0, -2.0, 0.0

Motion statements - POINT-TO-POINT MOTIONS

Two basic PTP motion commands: GOTO and GODLTA

The GOTO statement instructs the tool to go to a particular point location specified in the descriptive data. Two examples would be:

```
GOTO/P2  
GOTO/2.0, 7.0, 0.0
```

In the first statement, P2 is the destination of the tool point. In the second statement, the tool has been instructed to go to the location whose coordinates are $x = 2.0$, $y = 7.0$, and $z = 0$.

The GODLTA command specifies an incremental move for the tool. For example, the statement

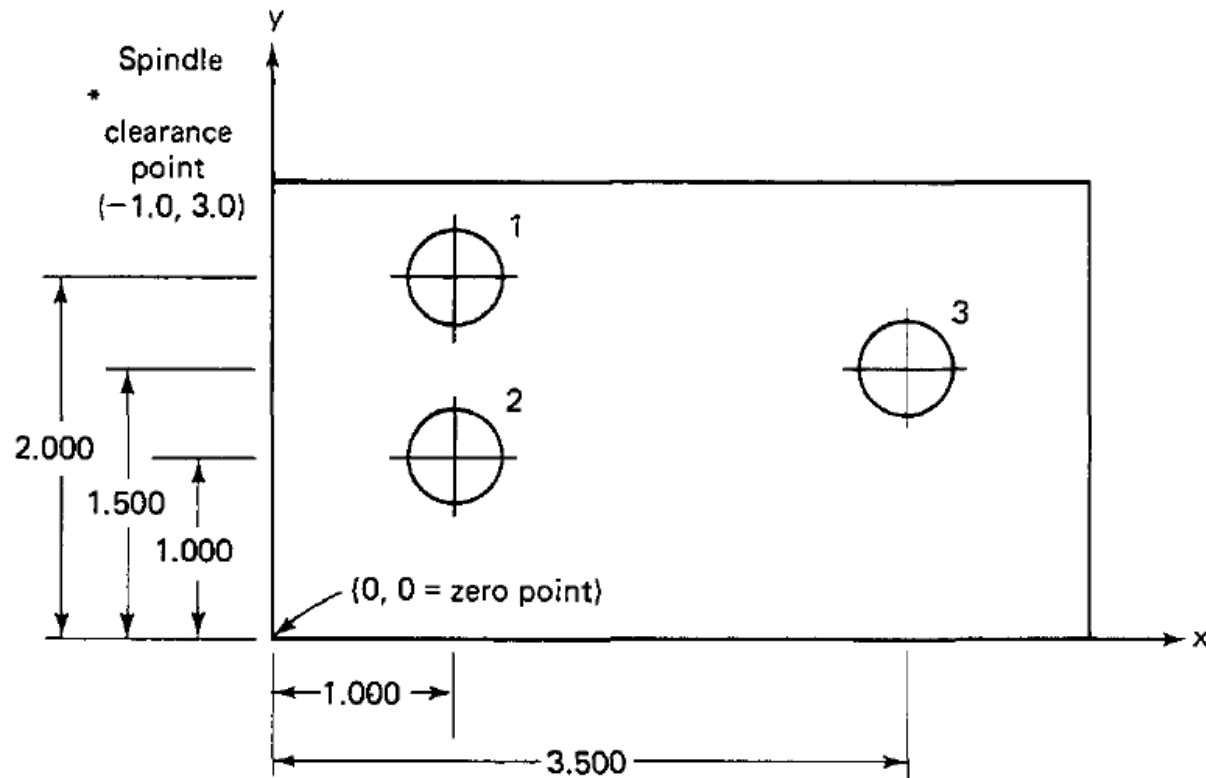
```
GODLTA/2.0, 7.0, 0.0
```

instructs the tool to move from its present position 2 in. in the x direction and 7 in. in the y direction. The z coordinate remains unchanged.

The GODLTA command is useful in drilling and related operations. The tool can be directed to a particular hole location with the GOTO statement. Then the GODLTA command would be used to drill the hole, as in the following sequence:

```
GOTO/P2  
GODLTA/0, 0, -1.5  
GODLTA/0, 0, +1.5
```

Example



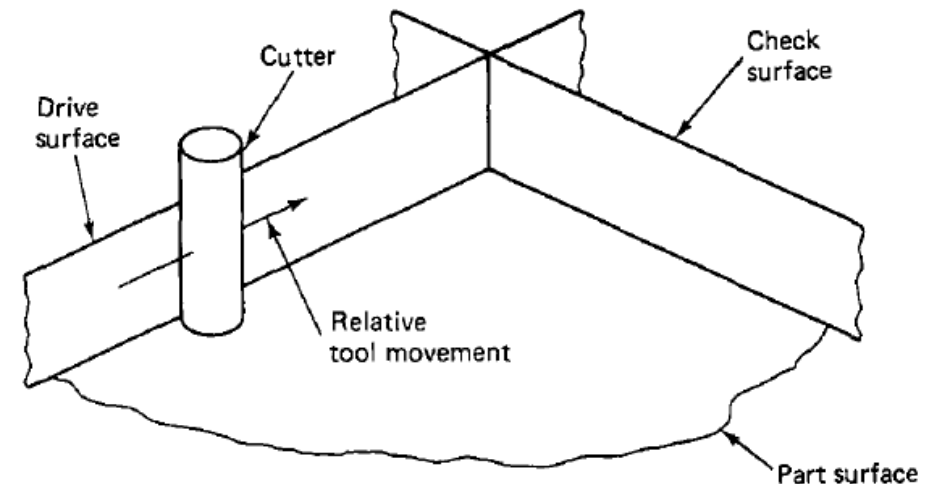
```
P1 = POINT/1.0, 2.0, 0
P2 = POINT/1.0, 1.0, 0
P3 = POINT/3.5, 1.5, 0
P0 = POINT/-1.0, 3.0, 2.0
FROM/P0
GOTO/P1
GODLTA/0, 0, -1.0
GODLTA/0, 0, +1.0
GOTO/P2
GODLTA/0, 0, -1.0
GODLTA/0, 0, +1.0
GOTO/P3
GODLTA/0, 0, -1.0
GODLTA/0, 0, +1.0
GOTO/P0
```

Contouring Motions

Contouring commands are somewhat more complicated because the tool's position must be continuously controlled throughout the move

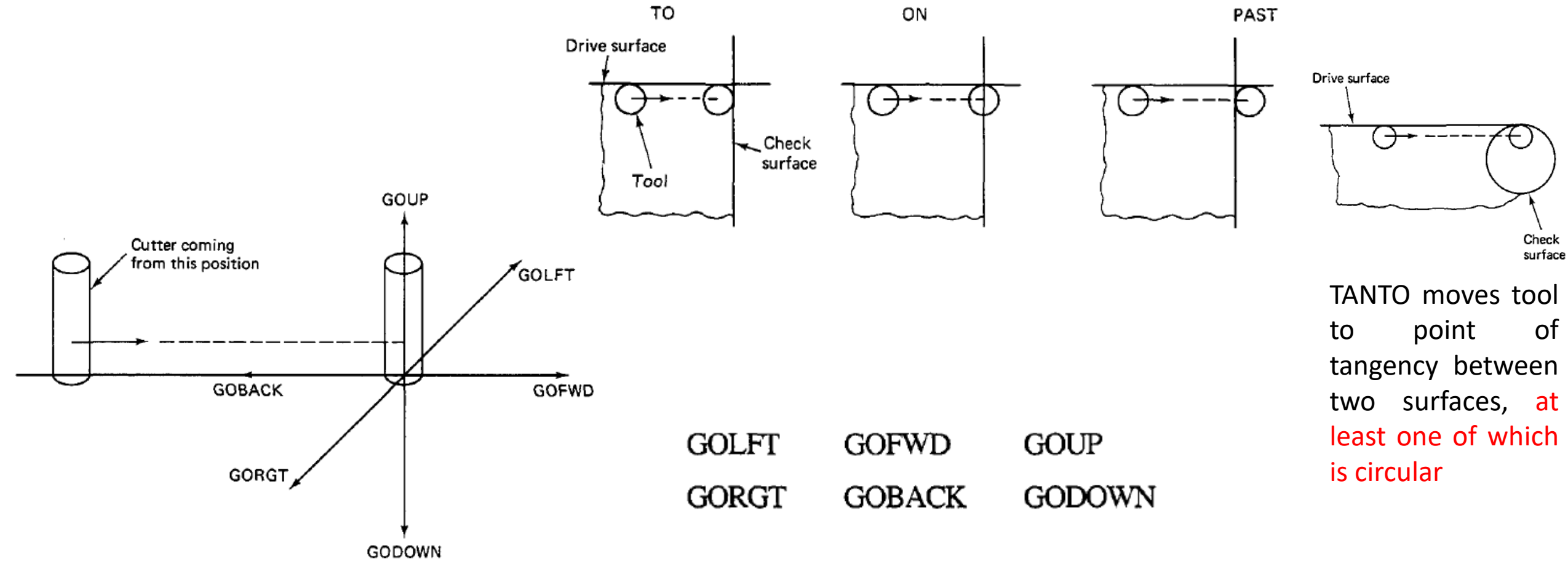
This control accomplish by intersecting surfaces

1. Drive surface - The surface guides the side of the cutter
2. Part surface - This is the surface on which the bottom of the cutter rides (may or may not be an actual surface of the workpart)
3. Check surface - This is the surface that stops the movement of the tool



APT contour motion statement

- The APT contour motion statement commands the cutter to move along the drive and part surfaces and the movement ends when the tool is at the check surface.
- The three **APT modifier words** are TO, ON, and PAST, and TANTO (when the drive surface is tangent to a circular check surface) their use with regard to the check surface is shown in figure



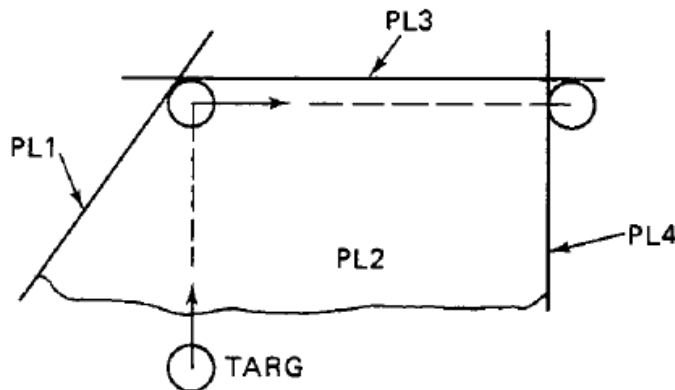
APT contour motion statement

FROM/TARG

GO/TO, PL1, TO, PL2, TO, PL3

GO command instructs the tool to move to the intersection of the *drive surface (PL1)*, the *part surface (PL2)*, and the *check surface (PL3)*. The periphery of the cutter is tangent to PL1 and PL3, and the bottom of the cutter is touching PL2

GO/TO Vs GOTO command : GOTO is used only for FTP motions and GO/TO is used to initialize the sequence of contouring motions



GORGT/PL3, PAST, PL4

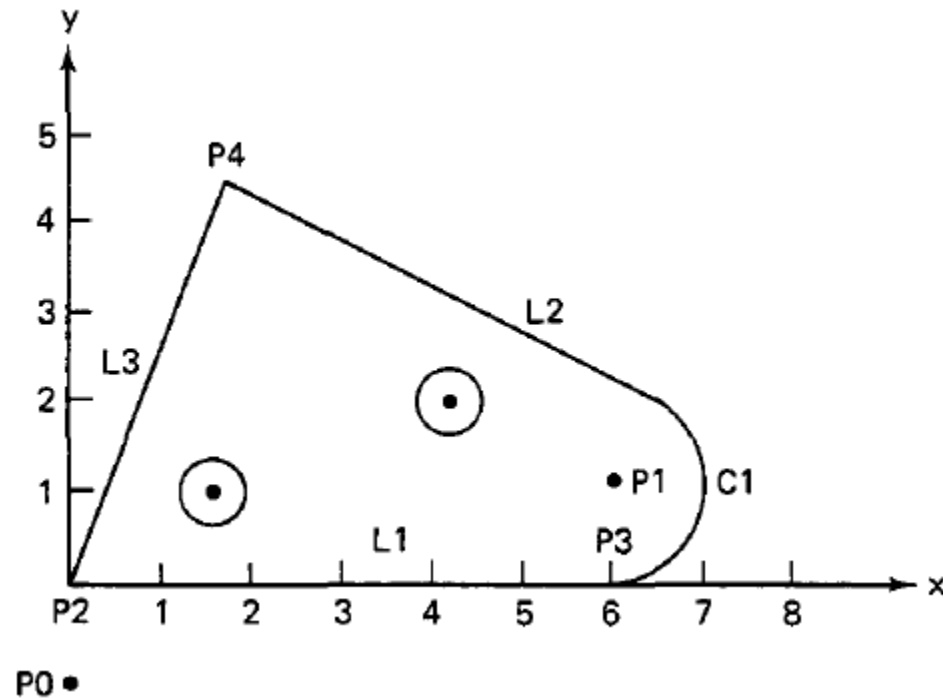
- The tool to move along PL3 (drive surface)
- The tool would continue until past surface PL4, which is the new check surface

or

GORGT/L3, PAST, L4

substituted lines L3 and L4 for planes PL3 and PL4, respectively

Example



FROM/P0
GO/TO, L1, TO, PL1, TO, L3
GORGT/L1, TANTO, C1
GOFWD/C1, PAST, L2
GOFWD/L2, PAST, L3
GOLFT/L3, PAST, L1
GOTO/P0

Postprocessor statements

Control the operation of the spindle, the feed, and other features of the machine tool called **postprocessor statements**

COOLNT/	RAPID
END	SPINDL/
FEDRAT/	TURRET/
MACHIN/	

Note:

with or without the slash (/).The statements without the slash are self-contained

Auxiliary statements

- CLPRNT
- CUTTER
- INTOL/
- OUTTOL/
- F1NI

CUTTER/.500

would instruct the APT program that the cutter diameter is 0.500 in

NC Positioning Systems

NC positioning system converts the coordinate axis values in the NC part program into relative positions of the tool and work part during processing.

Consider the simple positioning system –

- The system consists of a cutting tool and a worktable on which a work part is fixtured.
- The table is designed to move the part relative to the tool.
- The worktable moves linearly by means of a rotating leadscrew, which is driven by a stepper motor or servomotor

Two types of control systems are used in positioning systems:

(a) open loop

(b) closed loop

An open-loop system operates without verifying that the actual position achieved in the move is the same as the desired position.

A closed-loop system uses feedback measurements to confirm that the final position of the worktable is the location specified in the program.

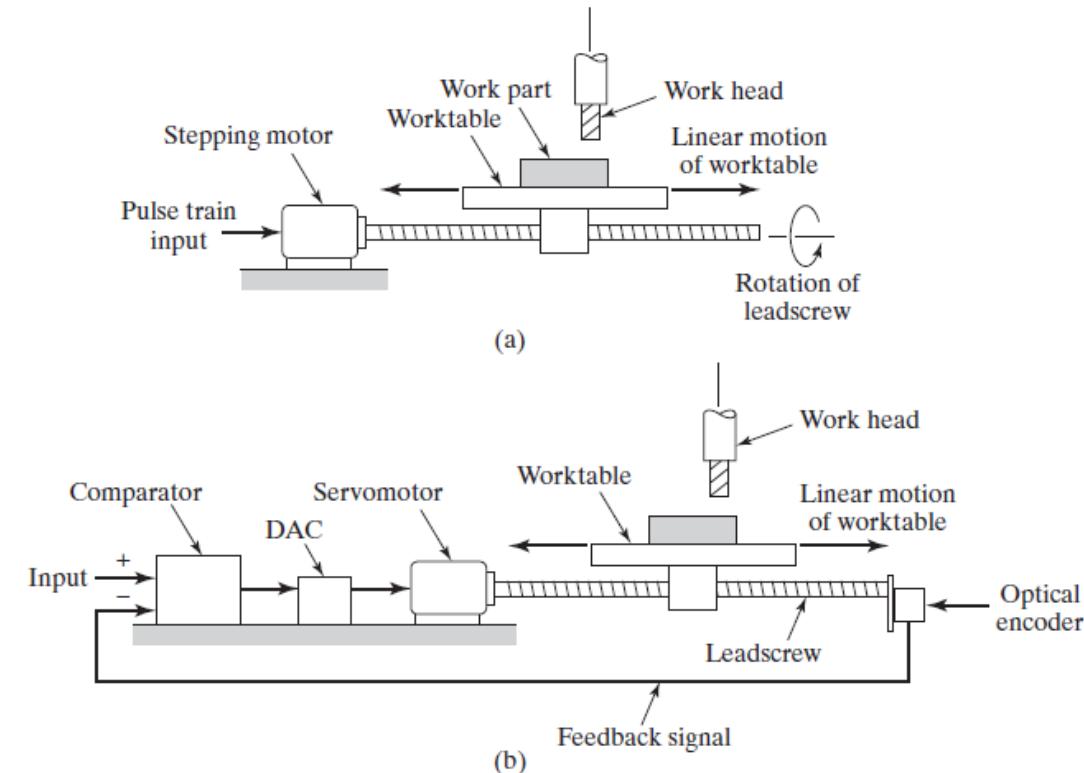
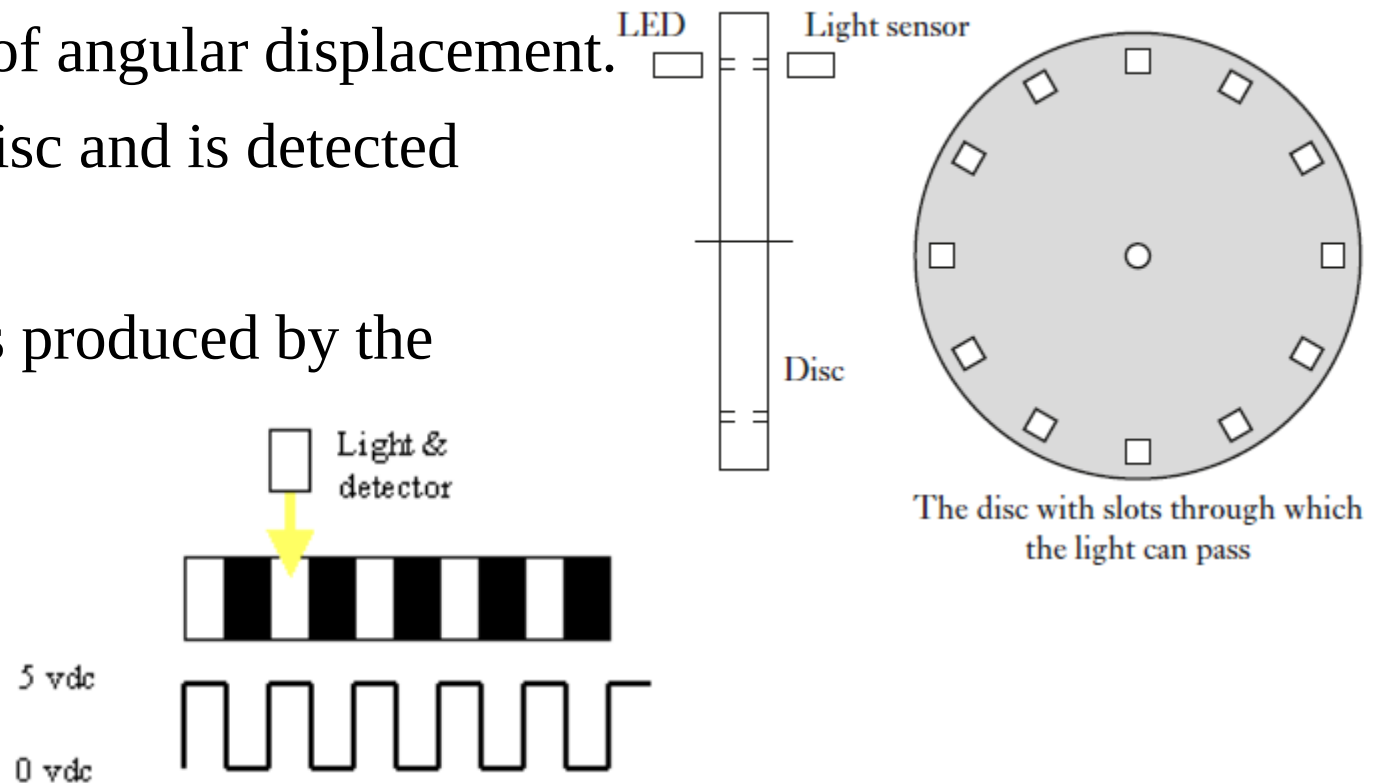


Figure 7.12 Two types of motion control in NC: (a) open loop and (b) closed loop Key: DAC = digital-to-analog converter.

Optical encoders

- An encoder is a device that provides a digital output as a result of a linear or angular displacement
- Encoders can be grouped into two categories: *incremental encoders and absolute encoders*
- Incremental encoders: detect **changes in position** (rotation) from some datum by generating pulses with each movement
- Absolute encoders : give the **actual angular position** at any given moment
- Incremental encoder for the measurement of angular displacement.
- A beam of light passes through slots in a disc and is detected by a suitable light sensor.
- When the disc is rotated, a pulsed output is produced by the sensor.

The number of pulses proportional to the angle through which the disc rotates



Optical encoders

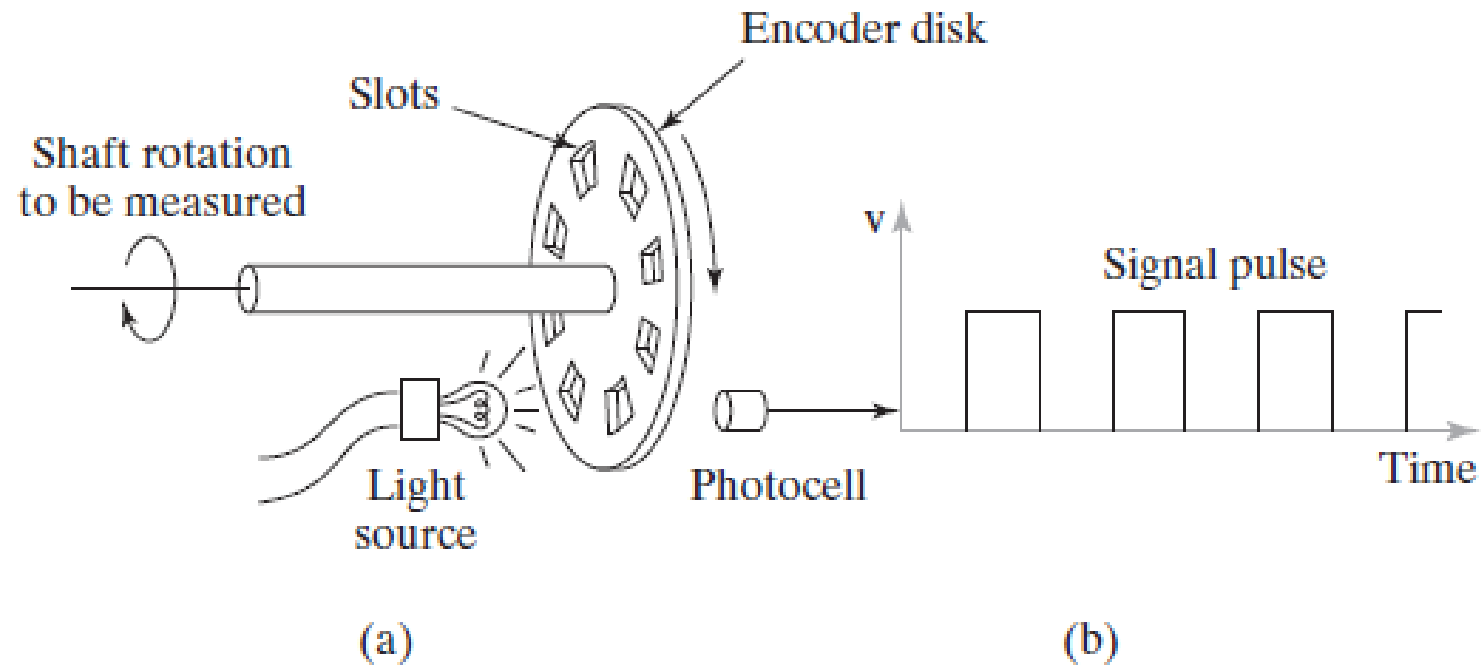
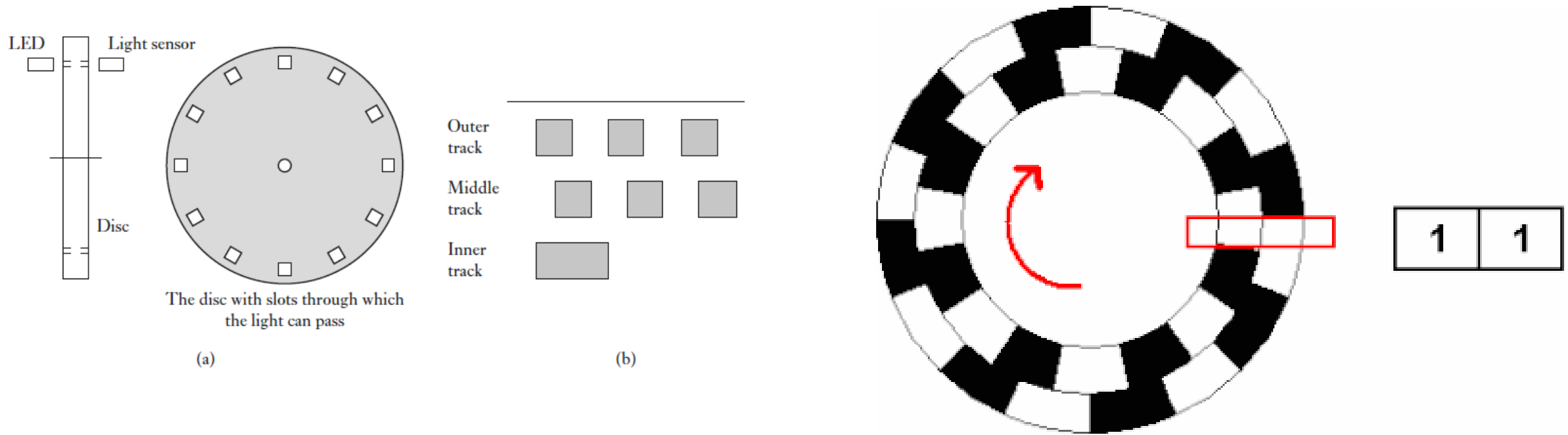


Figure 7.13 Optical encoder: (a) apparatus and (b) series of pulses emitted to measure rotation of disk.



- The outer and middle tracks have a series of equally spaced holes that go completely round the disc but with the holes in the middle track offset from the holes in the outer track by one-half the width of a hole.
- This offset enables the direction of rotation to be determined.
- In a **clockwise direction the pulses in the outer track lead those in the inner**
- In the anti-clockwise direction they lag

Precision in NC Positioning Systems



Control Resolution: This refers to the smallest possible movement a system can make.

- Imagine a ruler with millimeter markings. The control resolution would be 1 millimeter - the system can't move any smaller than that increment.
- Control resolution refers to the control system's ability to divide the total range of the axis movement into closely spaced points that can be distinguished by the Machine Control Unit (MCU)

Accuracy: This describes how close the system's actual position is to the commanded position. Ideally, the system should move exactly where it's instructed to. Accuracy considers any errors or deviations from the target.

Repeatability: This defines how consistently the system returns to the same position when repeatedly commanded to a specific location. It's about precision in achieving the same target point over and over.

CNC machining - zero positions

Machine Zero:

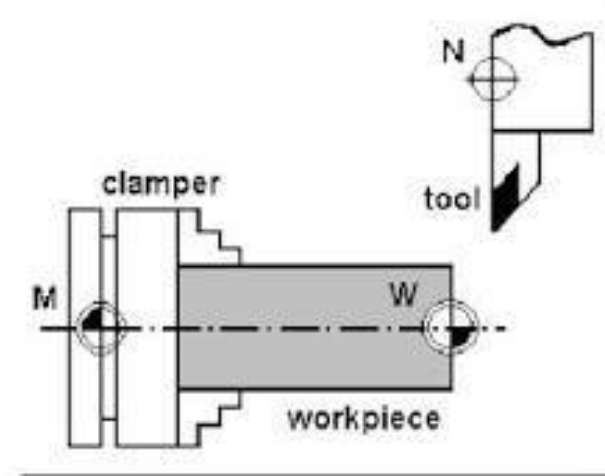
- This is a fixed reference point established by the manufacturer
- It's typically located at the farthest positive travel limits of the X, Y, and Z axes on the machine.
- This point is permanent and cannot be changed by the operator.

Workpiece Zero (or Program Zero):

- This is a user-defined reference point that corresponds to a specific location on the workpiece itself.
- This is the point you establish before running a CNC program to tell the machine where the part is located within the machine's overall coordinate system.

Zero Offset (Work Offset)

- This is the difference between the machine zero and the workpiece zero.
- It essentially tells the CNC controller **how much the programmed toolpath needs to be shifted relative to the machine's home position** to machine the part according to the program.



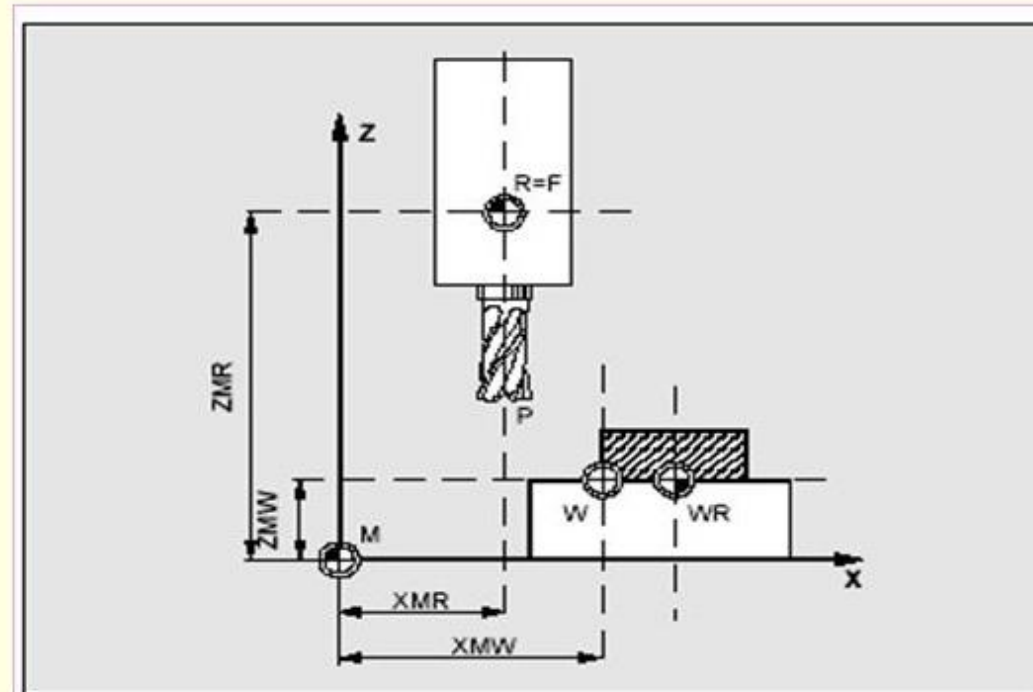
M = Machine zero point.
(unchangeable ref. Point
determined by machine
manufacturer)

N = Tool mounting
reference point

W = Work piece zero
point. (can be freely
determined by the
programmer and can be
moved within program)

CNC machining : Zero Offset

- Machine zero and work piece zero need not be identical.
- Depending on the type of work piece and the way it is clamped, the distance between the machine zero and work piece zero can vary
- In part program processing this is compensated by the zero offset



P - Tool setting point
W- Workpiece zero
F - Slide reference point
XMR, ZMR Reference point coordinates
XMW, ZMW Zero offset
M- Machine zero
R - Machine reference point
WR- Workpiece reference point

errors affects the accuracy of NC machine tools

Static errors

- Mechanical imperfections in the machine tool, such as misalignment of ways, wear on bearings, and backlash in gears.
- Thermal expansion of the machine tool due to changes in temperature.
- Tool wear and deflection.

Static errors can be minimized through proper calibration and maintenance of the machine tool, as well as by using tool wear compensation techniques.

errors affects the accuracy of NC machine tools

Dynamic errors are errors that are caused by the motion of the machine tool. They are typically more significant at higher speeds and accelerations.

Dynamic errors can be caused by factors such:

- The inertia of the machine tool components, which can cause the machine to lag behind the programmed toolpath.
- The compliance of the machine tool structure, which can cause the machine to deform under load.
- The control system dynamics, which can introduce delays and inaccuracies into the movement of the machine.

Dynamic errors can be minimized through careful design of the machine tool structure and control system, as well as by using feedforward control techniques.

- By understanding and minimizing both static and dynamic errors, manufacturers can ensure that their NC machine tools produce parts with high accuracy and precision

Basic Length Unit (BLU)

It represents the smallest increment of movement that a CNC machine tool can make in a given direction. In simpler terms, it's the minimum distance a CNC machine can move at one time.

$$\text{BLU} = \text{Lead screw pitch} / \text{Steps per motor revolution}$$

- Lead screw pitch: This is the distance the nut on the lead screw travels along the screw for each complete revolution
- Steps per motor revolution: This is the number of discrete angular movements a stepper motor can make in one complete revolution

In NC machine, the displacement length per one pulse output from machine is defined as a Basic Length Unit (BLU).

CNC computer each bit (binary digit) represents 1 BLU.

$$\text{Bit} = \text{BLU}$$

If one pulse makes a servo motor rotate by one degree and the servo motor moves the table by 0.0001 mm, one BLU will be 0.0001 mm.